AIR UNIVERSITY	Dept. of Cyber Security CS325 – Operating Systems **Semester Project** (CLO 2)	**Total Marks: 40** **Due Date** **Demo: 19-05-2026** **Report: 24-05-2026**

Instructions:

Please read the following instructions carefully before submitting the project.

- **The project will be submitted in groups of at most FOUR students.**
- **It should be clear that your project will not get any credit if:**
  - **The project is submitted after due date.**
  - **The code is generated by AI!**

Submission Procedure:

Submit all project reports (in PDF format) on Google Classroom. Demos will be conducted in the Lab.

Project title:

Modification of Linux OS kernel

Objectives:

The objective of this complex programming activity is to carry out research, investigation, analysis, design, and implementation of a real-world Operating Systems component that has the following attributes:

1. **Depth of Analysis Required**: The activity requires abstract thinking, originality in analysis to select suitable data structures and implementing standard algorithms.

2. **Interdependence**: The activity requires designing and building high level software that has many interdependent components or sub-parts such as development of a core OS module, integration of developed module into an existing OS kernel, and performance analysis of the developed module with the existing module.

3. **Innovation**: The activity involves creative use of OS principles and research-based knowledge in novel ways.

4. **Familiarity**: The activity can extend beyond previous experiences by applying novel OS approaches not studied in the course.

The subject-specific objectives include:

1. Understand the internal implementation of a key OS concept within the Linux kernel.
2. Modify the kernel source code related to the selected concept.
3. Recompile the kernel and configure the system to boot from the new kernel.

Guidelines:

- Members of each group should divide the tasks equally between them.
- While it is understood that one member cannot work on all aspects of the project, it is expected that each member should know the basic structure of the project.
- Code should be properly aligned and well-commented.
- The assessment shall be done on the basis of functionality (20 marks), report (10 marks) and demo (10 marks).

Project requirements:

In this project you are required to explore the Linux operating system at the kernel level by selecting a core operating system concept, identifying it in the kernel source and making a meaningful modification (method you have discussed in class) to its implementation in the kernel source code.

You will set up a development environment, identify and edit the relevant part of the Linux kernel source, recompile the kernel, and successfully boot into the modified version to observe and test the performance effects of the change.

The report must include the following sections:

- The steps to download, compile and boot from the compiled kernel (with screenshots)
- The process of identification of your selected kernel component within the source code
- Selection of an alternative methodology for modifying the chosen kernel component
- Criteria for evaluating the performance of kernel before and after modification, and discussion on results of comparative performance evaluation

You can select any kernel components to modify including, but not limited to, the following:

1. CPU scheduling algorithm
2. Page replacement algorithm
3. Page table implementation (hierarchical, hashed or inverted)